Harry Dawson

Curriculum Vitæ

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WORK EXPERIENCE

Postdoctoral Researcher (upcoming)
From 02/2027

Vilnius University, Lithuania

- Main project: uncovering new clues to the mystery of hot subdwarf stars.

Doctoral Researcher
04/2021 – Present

University of Potsdam

- Main project: volume-complete samples of hot subluminous stars in the *Gaia* era (supervisor: Prof. Dr. Dr. Stephan Geier).
- First author of a three-paper peer-reviewed series establishing and characterising this sample.
- Dedicated principal investigator on observing programmes at 2–8 m class facilities, from proposal writing through to data reduction and publication.

Teaching, Student Supervision and Mentoring
04/2021 – Present

University of Potsdam, Ondřejov Observatory, Czech Republic

- Project lead and co-organiser of an annual two-week research workshop on observational techniques at Ondřejov Observatory, Czech Republic, teaching M.Sc. students to observe with the 2 m-class Perek telescope. It involved supporting the students through photometric and spectroscopic observations, from planning the night through to reducing and analysing the resulting data.
- Co-supervised three M.Sc. students during my PhD, mentoring each on data acquisition, analysis, and final thesis write-up.
- Teaching assistant for the M.Sc.-level course *Stars and Stellar Evolution*, running tutorials and delivering the 1.5 hour lecture on several occasions.
- Instructor for a TOPCAT and ADQL data-mining workshop — building target lists from the *Gaia* archive.

Telescope Operator
05/2019 – 08/2020

University of Potsdam

- Alongside my Masters, I worked as a telescope operator and lab-course supervisor using the 50 cm OST observatory at the University of Potsdam.
- Maintained the upkeep of the telescope and taught the 2019/2020 generation of M.Sc. students to perform and optimise astronomical observations, and to reduce and analyse the acquired data.

EDUCATION

PhD in Astrophysics (thesis submitted)
04/2021 – Present

University of Potsdam, Germany

Master of Science in Astrophysics
10/2018 – 03/2021

University of Potsdam, Germany

Final grade: **1.2** — awarded on the German scale of 1.0 to 5.0, where 1.0 is the highest attainable mark.

Bachelor of Science in Physics and Mechanical Engineering
02/2013 – 10/2016

The University of Western Australia

Completed as a double major in physics and mechanical engineering.

RESEARCH PUBLICATIONS

A complete list of publications, including metrics, can be found in my **public ADS library**. It is also reproduced at the end of this document.

25 scientific contributions, including **20 peer-reviewed journal articles**; 3 as first author. Total citations: **193**. **h-index: 7**.

First author

- **Dawson, H.**, Dorsch, M., Munday, J., and 16 colleagues: *A 500 pc volume-limited sample of hot subluminous stars. III. The short-period binary population.* 2026arXiv260827673D **Accepted, A&A.**
- **Dawson, H.**, Dorsch, M., Geier, S., and 30 colleagues: *A 500 pc volume-limited sample of hot subluminous stars. II. Atmospheric*

parameters, mass distribution, and kinematics. 2026A&A...707A...6D **Cited: 10**

- **Dawson, H.**, Geier, S., Heber, U., and 25 colleagues: *A 500 pc volume-limited sample of hot subluminous stars. I. Space density, scale height, and population properties.* 2024A&A...686A..25D **Cited: 25**

TELESCOPE TIME

Total time awarded as principal investigator (PI) or dedicated principal investigator (dPI):

- **48 nights** at 2 m-class telescopes (INT / NOT).
- **37 nights** and **212 hours** at 4 m-class telescopes (SOAR / NTT / WHT / 3.5 m at Calar Alto).
- **81 hours** at 8 m-class telescopes (VLT X-shooter).

A further **84 hours** at 8 m-class telescopes (VLT UVES) as co-investigator. A full list is given at the end of this document.

TALKS, SEMINARS AND MEETINGS

20 contributed talks and posters since 2020, including **2 invited contributions**.

Invited

2025 Stars of Heidelberg stellar seminar, Heidelberg, Germany: *invited seminar.*
2025 12th Meeting on Hot Subdwarfs and Related Objects, Little Switzerland, USA: *invited conference opening.*

Selected contributed talks and posters

2026 AG Meeting, Garching, Germany: *talk*; EUROWD, Vienna, Austria: *poster.*
2025 EUROWD, Barcelona, Spain: *talk.*
2024 11th Meeting on Hot Subdwarfs and Related Objects, Armagh, UK: *talk.*
2022 10th Meeting on Hot Subdwarfs and Related Objects, Liège, Belgium; EUROWD, Tübingen; AG Meeting, Bremen: *talks.*
2021 Hot Subdwarf Meeting 9.5, Potsdam, Germany: *talk*; member of the local organising committee.
2020 AG Meeting, Potsdam, Germany: *poster.*

COLLABORATIONS AND ACADEMIC SERVICE

- Member of the **4MOST** Science Team, Milky Way Bulge and Disk Low Resolution Survey.
- Member of the variable stars working group, **BlackGEM** telescope array collaboration (2022–present).
- Member of the German Astronomical Society (*Astronomische Gesellschaft*) (2021–present).
- Referee for *The Astrophysical Journal* (ApJ).

TECHNICAL SKILLS

- Over 100 nights of observing experience across 1–8 m telescopes and instruments.
- Reduction of optical spectroscopic and photometric time-domain data (IRAF, PyRAF).
- Quantitative spectral analysis and model atmosphere fitting of hot stars.
- Spectral energy distribution fitting to derive radii, masses and reddening.
- Radial velocity extraction and orbital solutions for close binary systems.
- Light curve modelling of eclipses and ellipsoidal modulation.
- Handling and cross-matching large catalogues with ADQL, TOPCAT and *Gaia* archive.
- Modelling and correcting selection effects in volume-limited stellar samples.
- Galactic kinematics, orbit integration and population membership classification.
- Scientific Python (NumPy, SciPy, Astropy, Matplotlib); Linux, HPC (SLURM), git, LaTeX.
- Observing proposal writing as PI and dPI, from target selection to time justification.

Languages

English Native German Intermediate

REFERENCES

Prof. Dr. Dr. S. Geier
University of Potsdam, Germany
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Doctoral supervisor

Prof. Dr. U. Heber
University of Erlangen-Nürnberg, Bamberg, Germany
ulrich.heber@sternwarte.uni-erlangen.de
Co-supervisor

Dr. I. Pelisoli
University of Warwick, UK
Ingrid.Pelisoli@warwick.ac.uk
Mentor and collaborator

APPENDIX: FULL PUBLICATION LIST

25 scientific contributions in total, of which 20 are peer-reviewed journal articles. Complete and always-current record in my public ADS library.

First author

- **Dawson, H.**, Dorsch, M., Munday, J., and 16 colleagues: *A 500 pc volume-limited sample of hot subluminous stars. III. The short-period binary population.* 2026arXiv260827673D **Accepted, A&A.**
- **Dawson, H.**, Dorsch, M., Geier, S., and 30 colleagues: *A 500 pc volume-limited sample of hot subluminous stars. II. Atmospheric parameters, mass distribution, and kinematics.* 2026A&A...707A...6D **Cited: 10**
- **Dawson, H.**, Geier, S., Heber, U., and 25 colleagues: *A 500 pc volume-limited sample of hot subluminous stars. I. Space density, scale height, and population properties.* 2024A&A...686A..25D **Cited: 25**

Second and third author

- Geier, S., Dorsch, M., **Dawson, H.**, and 5 colleagues: *The first massive compact companion in a wide orbit around a hot subdwarf star.* 2023A&A...677A..11G **Cited: 13**

Contributing author

- Bhat, A., Dorsch, M., Heber, U., and 2 colleagues including **Dawson, H.**: *Intermediate-mass runaway and hypervelocity star candidates in DESI DR1.* 2026arXiv260830063B
- Roth, M. M., Roje, P., Vješnica, S., and 18 colleagues including **Dawson, H.**: *The ORCA-TWIN qCMOS experiment. I. Science case and commissioning at Calar Alto Observatory.* 2026AN....34770110R **Cited: 3**
- Deshmukh, K., Geier, S., Sana, H., and 7 colleagues including **Dawson, H.**: *Toward 3D orbits of wide sdO/B binaries. I. Six composite systems spatially resolved with VLTI/GRAVITY.* 2026A&A...712L..19D
- Munday, J., Rodriguez, A. C., Reindl, N., and 23 colleagues including **Dawson, H.**: *V407 Vul: a triple star system with an AM CVn detectable by gravitational wave observatories.* 2026arXiv260725557M
- Pritzkuleit, M., Dorsch, M., Miller Bertolami, M. M., and 3 colleagues including **Dawson, H.**: *Stars on the ascending helium giant branch. I. From white dwarf merger to helium giant: the evolutionary state of the rapidly rotating hot subdwarf HE 1518−0948.* 2026A&A...710A..99P **Cited: 1**
- Heber, U., Kufleitner, L., Dorsch, M., and 4 colleagues including **Dawson, H.**: *Hot subdwarf stars from the Hamburg Quasar Survey.* 2026A&A...708A.115H **Cited: 4**
- Mattig, F., Barlow, B. N., Liu, D., and 9 colleagues including **Dawson, H.**: *The massive hot subdwarf binary LAMOST J065816.72+094343.1.* 2026A&A...706L...2M
- Culpan, R., Dorsch, M., Pelisoli, I., and 15 colleagues including **Dawson, H.**: *A search for missing binaries: blue horizontal-branch stars in binary systems in the inner Galactic halo.* 2025A&A...704A.326C **Cited: 4**
- Kurpas, M., Dorsch, M., Geier, S., and 12 colleagues including **Dawson, H.**: *The twin red giant branch system BD+20 5391: a case study of low-mass double-core evolution.* 2025A&A...702A.200K **Cited: 4**
- Munday, J., Pelisoli, I., Tremblay, P.-E., and 14 colleagues including **Dawson, H.**: *The DBL survey II: towards a mass–period distribution of double white dwarf binaries.* 2025MNRAS.541.3494M **Cited: 5**
- Bhat, A., Dorsch, M., Geier, S., and 5 colleagues including **Dawson, H.**: *Ancient, eclipsing, tidally locked: a blue lurker progenitor in the population of extreme-velocity star candidates.* 2025A&A...700L..23B **Cited: 1**
- Krzesinski, J., Uzundag, M., Kumari, G. A., and 9 colleagues including **Dawson, H.**: *A comprehensive search for hot subdwarf stars using Gaia and TESS. II. Uncovering new pulsators and close binary systems.* 2025A&A...700A..71K **Cited: 2**
- Munday, J., Pelisoli, I., Tremblay, P.-E., and 8 colleagues including **Dawson, H.**: *Correction to: The DBL survey I: discovery of 34 double-lined double white dwarf binaries.* 2025MNRAS.540.1272M **Cited: 1**
- Groot, P. J., Bloemen, S., Vreeswijk, P. M., and 76 colleagues including **Dawson, H.**: *The BlackGEM telescope array. I. Overview.* 2024PASP..136k5003G **Cited: 49**
- Munday, J., Pelisoli, I., Tremblay, P.-E., and 8 colleagues including **Dawson, H.**: *The DBL survey I: discovery of 34 double-lined double white dwarf binaries.* 2024MNRAS.532.2534M **Cited: 30**
- Uzundag, M., Krzesinski, J., Pelisoli, I., and 5 colleagues including **Dawson, H.**: *A comprehensive search for hot subdwarf stars using Gaia and TESS. I. Pulsating hot subdwarf B stars.* 2024A&A...684A.118U **Cited: 20**
- Reindl, N., Islami, R., Werner, K., and 10 colleagues including **Dawson, H.**: *The bright blue side of the night sky: spectroscopic survey of bright and hot (pre-)white dwarfs.* 2023A&A...677A..29R **Cited: 22**

Data catalogues, observing proposals and circulars

- Malesani, D. B., Sánchez-Sierras, J., Martín-Carrillo, A., and 6 colleagues including **Dawson, H.**: *EP250421a: NOT optical observations.* GCN Circular. 2025GCN.40207....1M
- Dorsch, M., Battich, T., Bauer, E., and 20 colleagues including **Dawson, H.**: *A far-UV treasury survey for hot subdwarf stars.* HST observing proposal. 2024hst..prop17697D
- **Dawson, H.**, Geier, S., Heber, U., and 25 colleagues: *VizieR online data catalogue: hot subluminous stars 500 pc volume-limited sample (Dawson+, 2024).* 2024yCat..36860025D
- Uzundag, M., Krzesinski, J., Pelisoli, I., and 5 colleagues including **Dawson, H.**: *VizieR online data catalogue: pulsating hot subdwarf B stars (Uzundag+, 2024).* 2024yCat..36840118U

APPENDIX: SUCCESSFUL TELESCOPE APPLICATIONS

Listed as principal investigator (PI) first, then as dedicated principal investigator (dPI), then as co-investigator (Co-I).

As principal investigator

NTT / EFOSC2 — 40.3 h
Unveiling the hidden population of hot pre-helium white dwarfs.
ESO Period P117 — 117.29U6. PI: H. Dawson.

As dedicated principal investigator

VLT / X-shooter — 34 h 37 m
Stability criterion for mass transfer on the first giant branch.
ESO Period P116 — 116.28ZZ.002 (run 2). PI: M. Vučković; dPI: H. Dawson.

VLT / X-shooter — 46 h 13 m
Stability criterion for mass transfer on the first giant branch.
ESO Period P117 — 117.2AET.002 (run 2). PI: M. Vučković; dPI: H. Dawson.

WHT / ISIS — 49.1 h (filler)
Northern-Cone spectroscopic survey of hot subdwarf stars.
ING Semester 2027A — SW2026b08 (WS2026B2-008). PI: I. Pelisoli; dPI: H. Dawson.

SOAR / Goodman — 22 nights
Radial velocity variability follow-up in the 500 pc sample of hot subdwarfs.
Semesters 2023B–2026A, 930 M2 setup — 2023B-711595 (5 n), 2024B-520098 (4 n), 2025A-537257 (5 n), 2025B-972608 (4 n), 2026A-730421 (4 n). PI: M. Vučković; dPI: H. Dawson.

CAHA 3.5 m / PMAS — 15 nights
Northern-Cone spectroscopic survey of hot subdwarfs: a volume-limited approach.
GTO time. PI: A. Soemitro; dPI: H. Dawson.

WHT / ISIS — 70 h (filler)
Northern-Cone spectroscopic survey of hot subdwarfs: a volume-limited approach.
CAT Semester 2025A. PI: R. Raddi; dPI: H. Dawson.

WHT / ISIS — 52.3 h (filler)
Northern-Cone spectroscopic survey of hot subdwarfs: a volume-limited approach.
ING Semester 2025A. PI: I. Pelisoli; dPI: H. Dawson.

NOT / ALFOSC — 6 nights
Unveiling the hidden population of hot pre-helium white dwarfs.
Programme 73-202. PI: R. Raddi; dPI: H. Dawson.

NOT / ALFOSC — 6 nights
Hunting for radial velocity variability in the 500 pc sample of hot subdwarfs.
Programme 71-201. PI: R. Raddi; dPI: H. Dawson.

INT / IDS — 11 nights
Unveiling the nature of the companions to hot subdwarf stars within 500 pc.
ING Semester 2024A — ING.NL.24A.005. PI: I. Pelisoli; dPI: H. Dawson.

INT / IDS — 10 nights
Constraining the binary population of hot subdwarf stars in the Gaia era.
ING Semester 2023B — ING.NL.23B.002. PI: I. Pelisoli; dPI: H. Dawson.

INT / IDS — 10 nights
Constraining the binary population of hot subdwarf stars in the Gaia era.
ING Semester 2023B — ING.NL.23B.005. PI: J. Munday; dPI: H. Dawson.

INT / IDS — 5 nights
A volume-limited sample of hot subdwarf stars within 500 pc.
ING Semester 2022B — P5 / ING.NL.22B.005. PI: I. Pelisoli; dPI: H. Dawson.

Mercator 1.2 m / HERMES — 110 h
Constraining the close binary fraction within 500 pc.
2025–2026. Monitoring of sdB+WD binaries in the northern 500 pc sample. PI: K. Deshmukh; dPI: H. Dawson.

ANU 2.3 m / WiFeS — 5 h GTO + 20 h open

Southern Galactic cone sample of hot subdwarf stars.
PI: N. Rodriguez-Segovia; dPI: H. Dawson.

ANU 2.3 m / WiFeS — 5 h GTO

Companions of thermonuclear supernovae.
PI: N. Rodriguez-Segovia; dPI: H. Dawson.

As co-investigator

VLT / UVES — 59 h 15 m

Stability criterion for mass transfer on the first giant branch.
ESO Period P117 — 117.2AET.002 (run 1; run 2 above as dPI). PI: M. Vučković.

VLT / UVES — 24 h 22 m

Stability criterion for mass transfer on the first giant branch.
ESO Period P116 — 116.28ZZ.002 (run 1; run 2 above as dPI). PI: M. Vučković.

VLTI / GRAVITY — 6.0 h

First dynamical mass measurements for six benchmark long-period sdO/B binaries.
ESO Period FTC 117A — 117.2AW4. PI: K. Deshmukh.

VLTI / GRAVITY — 6.0 h

Do some hot subdwarf binaries descend from hierarchical triples?
ESO Period P116 — 116.28ZD. PI: K. Deshmukh.

VLT / UVES — 3.1 h

The eclipsing sdB+WD binary SDSS J070215.20+121753.4.
ESO Period FTC 117A — 117.2AXC. PI: F. Mattig.

NTT / EFOSC2 — 27.4 h

An analysis of hot and bloated pre-ELMs in close binary systems.
ESO Period P114 — 114.270C. PI: S. Geier.

NTT / ULTRACAM — 28.5 h

A comprehensive search for pulsating hot subdwarf B stars using a Gaia volume-limited sample.
ESO Period P113 — 113.26FE. PI: M. Uzundag.